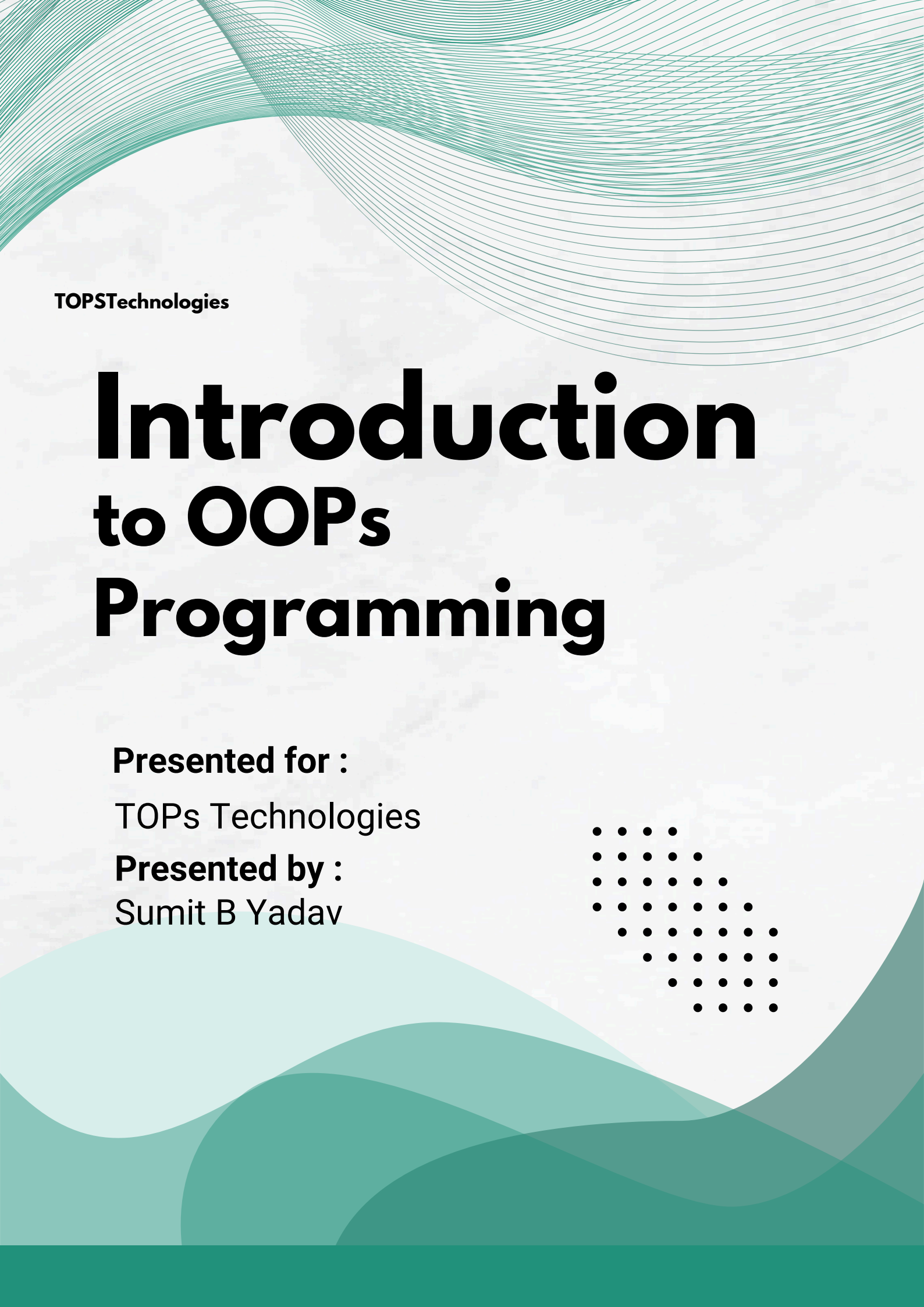




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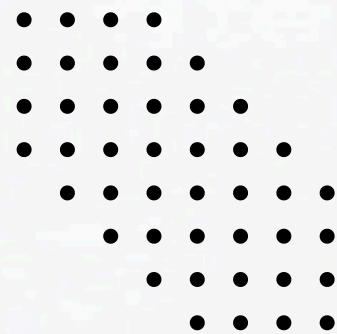
Introduction to OOPs Programming

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Basic Structure of a C++ Program

Que 1

In C++, a function is a block of code that performs a specific task. Functions allow you to group statements together to perform a particular job, and they can be reused multiple times throughout a program. Functions are essential for organizing code, reducing redundancy, and improving readability and maintainability.

Components of a Function

In C++, a function has three main components:

- 1. Function Declaration (or Function Prototype)*
- 2. Function Definition*
- 3. Function Calling*

1. Function Declaration

A function declaration tells the compiler about the function's name, return type, and parameters (if any), but it does not provide the implementation. It is typically placed at the beginning of a program or before the main() function, so the compiler knows about the function before it is used.

2. Function Definition

A function definition provides the actual implementation of the function. It specifies what the function does when it is called. The function definition includes the same signature as the declaration but also includes the body of the function.

Basic Structure of a C++ Program

3. Function Calling

A function call is an expression that invokes the function to perform its task. When a function is called, the program control is transferred to the function, and the function's statements are executed. Once the function finishes executing, control is returned to the point where the function was called.

Que 2

The scope of a variable refers to the region of the program where that variable is accessible or valid. The scope of a variable is determined by where it is declared, and it defines the lifetime and visibility of that variable within the program.

1. Local Scope

A local variable is a variable that is declared inside a function, block, or any specific code block (e.g., loops, conditionals). Its scope is limited to that function or block, meaning it can only be accessed or modified within that region.

2. Global Scope

A global variable is a variable that is declared outside of all functions, typically at the top of the program. Its scope is throughout the entire program, meaning it is accessible from any function, as long as it is not shadowed by a local variable with the same name.

Basic Structure of a C++ Program

Do-while Loop

A do-while loop is similar to the while loop, but with one key difference: the condition is checked after the loop body is executed. This guarantees that the loop will run at least once, even if the condition is false.

Que 3

Recursion in C++ is a programming technique in which a function calls itself directly or indirectly to solve a problem. Each recursive call solves a smaller portion of the problem, and when the base condition is met, the function stops calling itself and begins returning values back to the previous calls.

In recursive functions, two key elements are crucial:

1. **Base Case:** The condition that stops the recursion and returns the result to the previous function call.
2. **Recursive Case:** The part of the function that calls itself with a modified argument to solve a smaller instance of the problem.

A common example of recursion is calculating the factorial of a number. The factorial of a number n is the product of all positive integers from 1 to n :

$$n! = n \times (n-1) \times (n-2) \times \dots \times 1$$

For example:

- $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$
 $0! = 1$ (by definition, the factorial of 0 is 1)
- $10! = 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 3628800$

Basic Structure of a C++ Program

```
#include <iostream>
using namespace std;

// Recursive function to calculate the factorial of a number
int factorial(int n) {
    // Base case: if n is 0, return 1
    if (n == 0) {
        return 1;
    }
    // Recursive case: n * factorial(n - 1)
    else {
        return n * factorial(n - 1);
    }
}

int main() {
    int num;

    cout << "Enter a number to calculate its factorial: ";
    cin >> num;

    // Calling the factorial function
    int result = factorial(num);

    cout << "The factorial of " << num << " is: " << result << endl;

    return 0;
}
```

Basic Structure of a C++ Program

Que 4

A function in C++ is a declaration of a function that provides information about the function's return type, its name, and the types of its parameters (but without the body of the function). It acts as a forward declaration, allowing the function to be used before its full definition appears in the code.

Function Declaration Before Use:

- C++ requires that the compiler knows about the function's signature (name, return type, parameters) before it is called in the program.
- If you call a function before its definition appears in the code, a function prototype ensures that the compiler knows the function's signature, avoiding errors.

Improves Readability and Organization:

- By separating the function declaration (prototype) from the function definition, the code can be better organized, especially in large programs with multiple functions. This separation allows the function definitions to be placed after the main() function, for instance, or even in separate source files.

Function help the compiler to check the function calls for correctness. The number and type of arguments used in the function call are verified against the prototype. If they do not match, the compiler will generate an error or warning.